

Yaswanth Muppalla

Atlanta, GA | +1 845-663-4330 | yaswanthmuppalla2025@gmail.com | [LinkedIn](#) | [GitHub](#)

AI Platform Engineer | Senior Software Engineer | Backend Systems | AWS | Data Engineering

PROFESSIONAL EXPERIENCE

American Express | Senior Software AI Engineer

Apr 2023 - Present

Impact: Built Java/Flink, Airflow, and IRIS-backed RAG systems for credit oversight workflows, improving reporting efficiency by 35%, achieving 99% data accuracy, reducing rerun/debug cycles by 3x, and accelerating AI copilot support.

- Partnered with BCA/CCA data pipeline teams and built JSON-configurable Apache Flink pipelines for ingestion, transformation, joins, validation, and downstream publishing, translating workflow knowledge into AI-ready pipeline patterns for credit oversight use cases.
- Automated Airflow DAG orchestration for credit data processing with dependency management, rerun handling, SLA monitoring, data-quality checks, and failure alerting, creating reliable operational signals for downstream AI copilot and RAG workflows.
- Designed an IRIS-backed RAG ingestion pipeline with document extraction, deduplication, upsert handling, segmentation, chunking, metadata enrichment, and batch publishing for semantic retrieval.
- Developed a RAG agent harness that combined IRIS semantic search, scoped retrieval, prompt assembly, source-ledger tracking, and grounded response generation for credit oversight users.
- Built a context harness for chatbot-style interactions, managing session context, selected documents, retrieved chunks, policy context, user query intent, and controlled LLM input assembly.
- Added observability and control layers across the AI workflow, including retrieval logs, execution traces, citation mapping, guardrails, cache behavior, and audit-friendly response tracking.

Skechers | Software Engineer

Oct 2020 - Mar 2023

Impact: Delivered 15+ Python APIs and event-driven services, maintained 99.9% uptime, reduced costs by 22%, migrated 35M+ records, and cut incident resolution time by 40%.

- Developed 15+ Python/Pydantic REST APIs on microservices-style, event-driven AWS Lambda/Step Functions, sustaining 99.9% uptime and cutting costs 22%.
- Migrated 35M+ SFMC records into MongoDB and improved query performance by 50% through indexing, caching, and data-access optimization.
- Integrated SessionM, SFMC, MongoDB, and AWS serverless services for loyalty/marketing workflows and sub-500ms transactions.
- Applied event-driven API architecture across Lambda, Step Functions, validation, orchestration, and data-access layers.
- Improved production resilience with CloudWatch/Datadog alerts, API failure tracing, retry handling, and SLA-focused incident feedback.

SOCH Inc. | Full-Stack Developer - Richmond, VA

Jan 2020 - Sep 2020

- Built a Spring Boot ticketing system with REST microservices and full-stack workflow screens for government clients.
- Used Java ExecutorService/CompletableFuture for async ticket updates, notifications, and background job handling.
- Managed concurrency with synchronized services, thread-safe collections, timeouts, and Spring Boot exception handling.

FOUNDER-LED TECHNICAL PROJECTS

OmInsights - Founder-Led AI Platform Build

Jan 2024 - Present

Impact: Built a 6-component AI platform with tenant-scoped RAG, Context Engine orchestration, MCP tool execution, async file processing, and source-backed answers over owned documents.

- Architected Angular UI, FastAPI Business Service, Context Engine, MCP Server, File Processing worker, and AWS-backed deployment paths with CI/CD and tenant-aware AWS operations.
- Built document intelligence pipelines for upload, parse, normalize, chunk, embed, persist, retrieve, and cite flows for grounded RAG responses with source ledgers and retrieval quality controls.
- Designed Context Engine flows for chat/session memory, selected context, KB injection, prompt assembly, and LLM/context harness design for multi-turn RAG orchestration and scoped answer grounding.
- Implemented MCP/API runtime patterns with tool catalog APIs, native/template tools, scoped document search, execution logging, cache behavior, and access-aware tool selection.
- Combined source-ledger, citation, guardrail, caching, and semantic caching patterns to keep AI outputs grounded and traceable.

TECHNICAL SKILLS

AI/LLM Systems: RAG, embeddings, vector search, LangChain, MCP/FastMCP, OpenAI, Amazon Bedrock, prompt engineering, LLMops, semantic caching, context orchestration

Backend: Python, FastAPI, Pydantic, Java 17, Spring Boot, REST APIs, JWT, microservices, Java concurrency, ExecutorService, CompletableFuture, TypeScript, Angular

Cloud & Data: AWS Lambda, ECS, S3, SQS, Step Functions, API Gateway, EventBridge, Docker, CI/CD, MongoDB, Redis, MemSQL, Kafka, Apache Flink, Airflow

Retrieval & Storage: OpenSearch Vector DB, MongoDB, document chunking, source ledgers, searchable chunk publishing, Elasticsearch

Observability: Datadog, CloudWatch, Kibana, X-Ray, ELK Stack, structured logging, SLA alerts, failure tracing

EDUCATION

State University of New York, New Paltz | Master of Science, Computer Science | CGPA: 3.6/4.0

May 2020

JNTUK | Bachelor of Technology | CGPA: 8.5/10

2018